

Implementing Security Monitoring and Logging (4e)

Fundamentals of Information Systems Security, Fourth Edition - Lab 08

Student:

Chidima Onyenobi

Email:

conyenobig@hc.edu

Time on Task:

2 hours, 44 minutes

Progress:

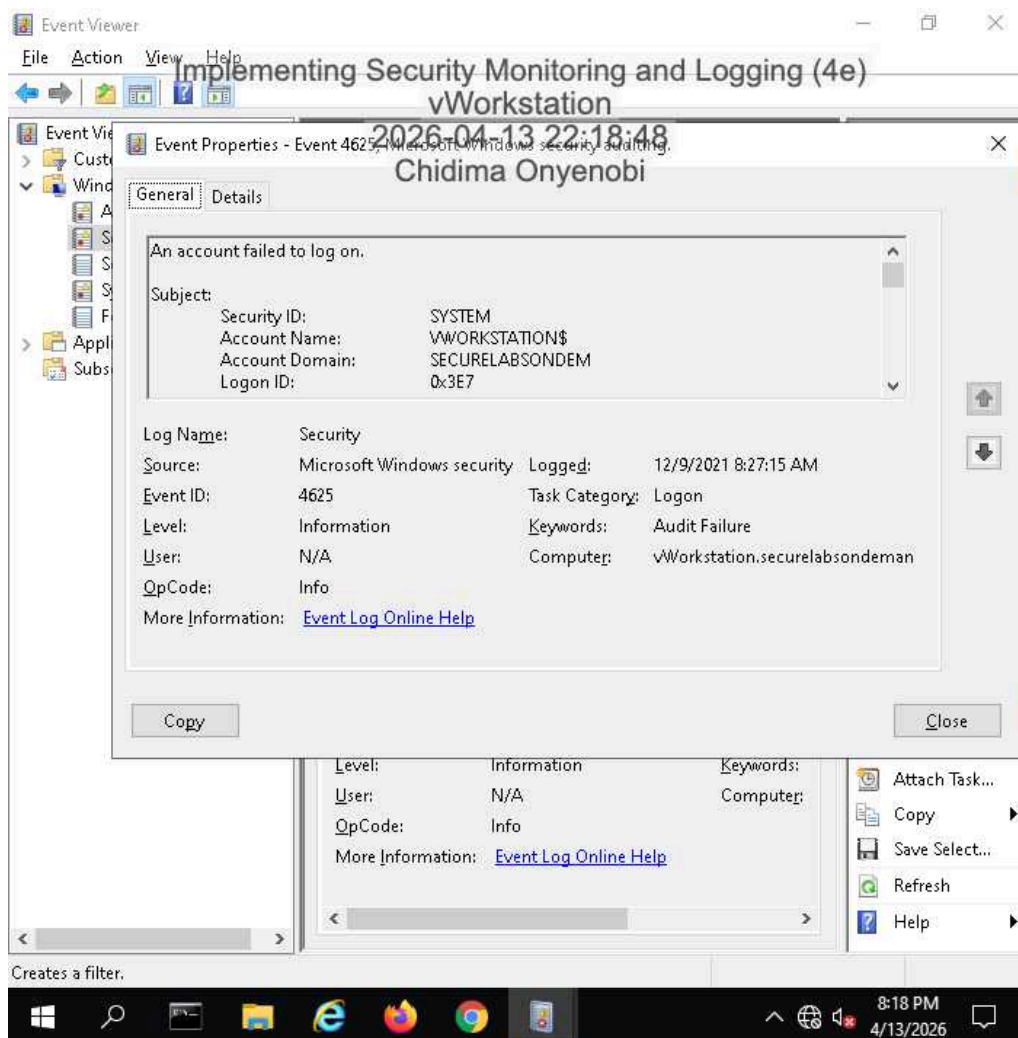
100%

Report Generated: Tuesday, April 14, 2026 at 1:50 AM

Hands-On Demonstration

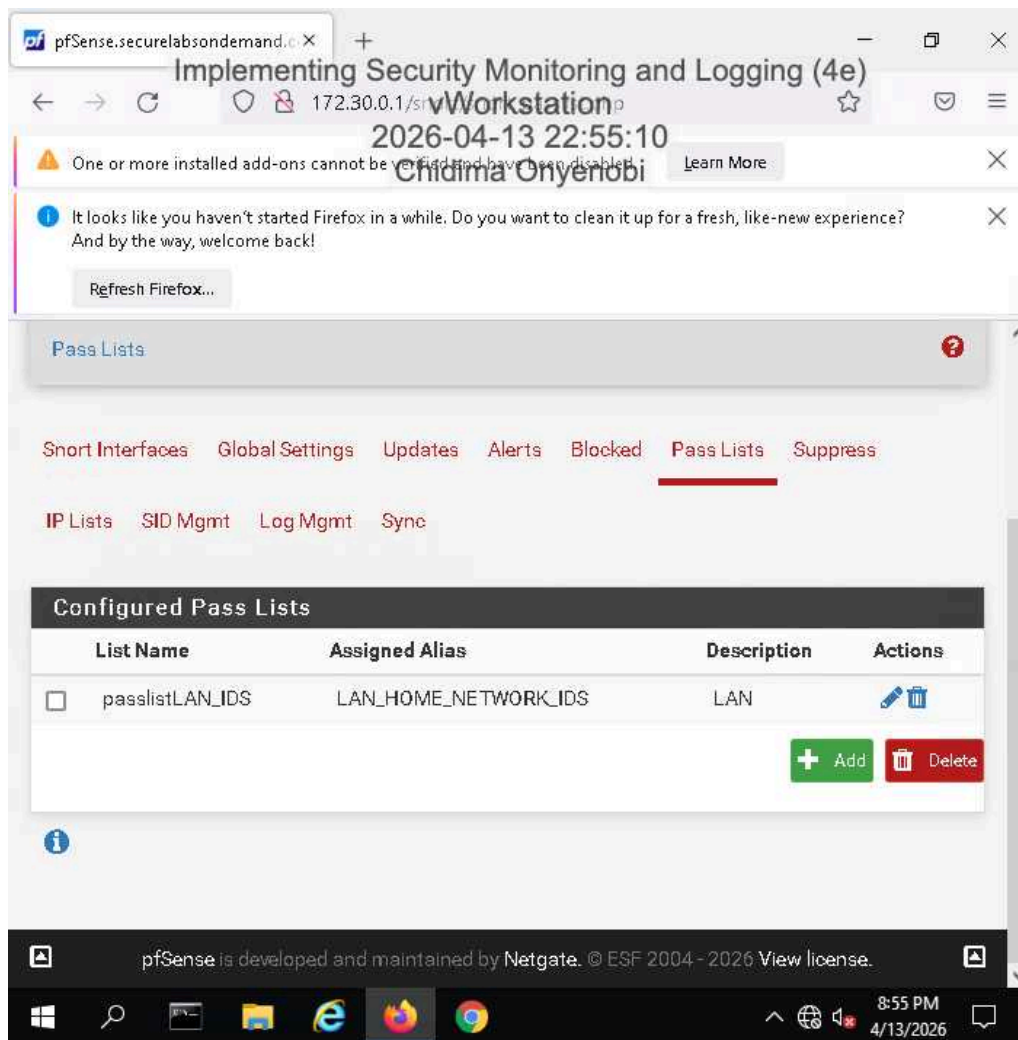
Part 1: Identifying Failed Logon Attempts on Windows Systems

8. Make a screen capture showing the **Security Event Properties** dialog box on the **vWorkstation**.

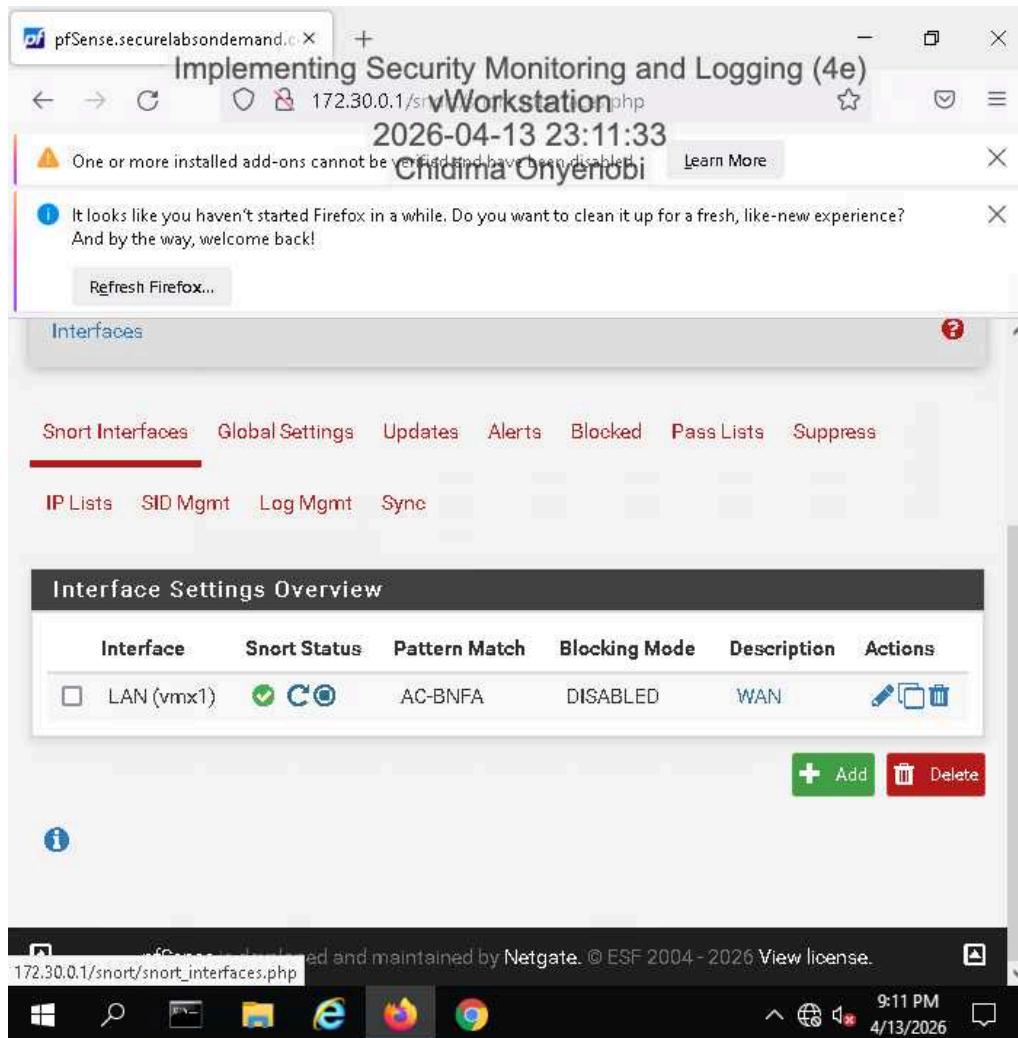


Part 2: Monitoring Network Activity with Snort

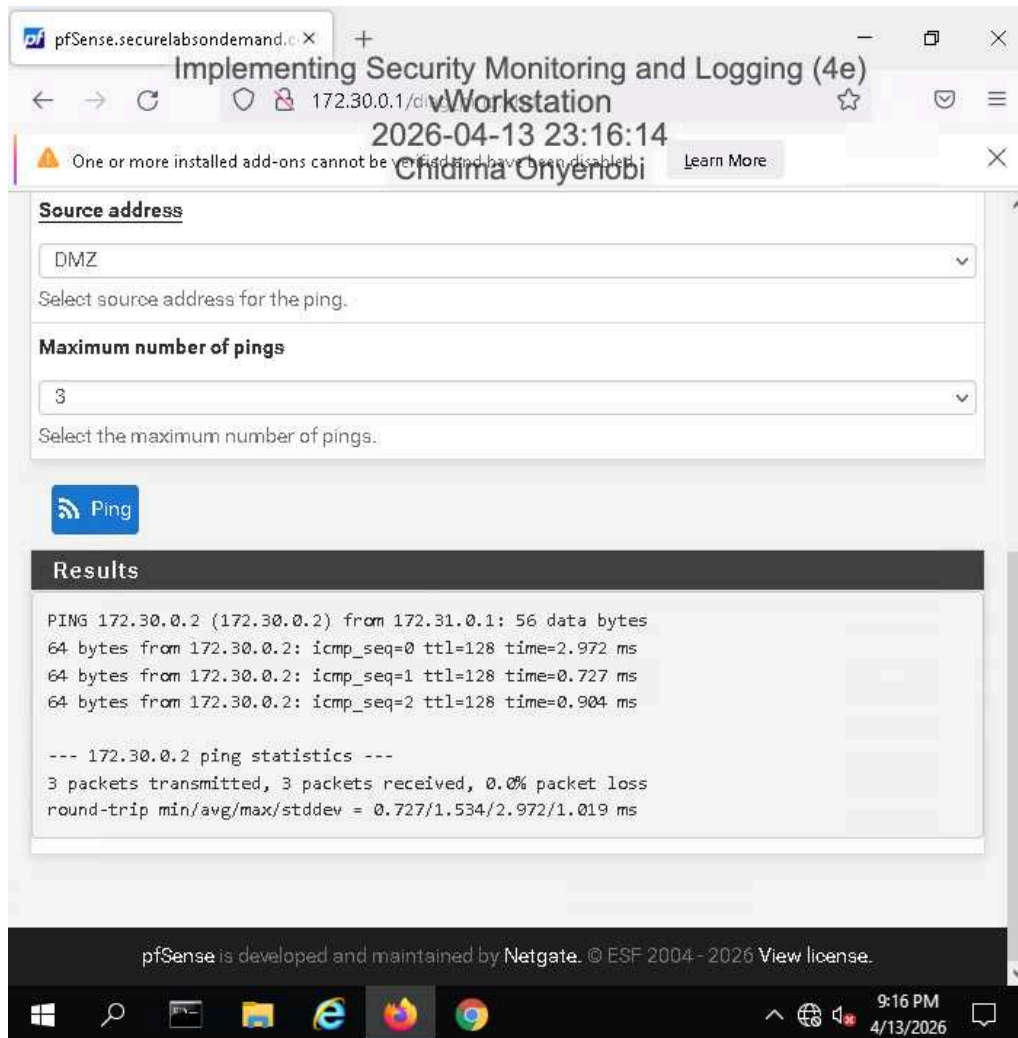
17. Make a screen capture showing the updated Pass Lists page.



31. Make a screen capture showing the active Snort status on the LAN interface.



36. Make a screen capture showing the successful ping results.



Implementing Security Monitoring and Logging (4e)

Fundamentals of Information Systems Security, Fourth Edition - Lab 08

41. Make a screen capture showing the ICMP alerts in the Snort Active Log.

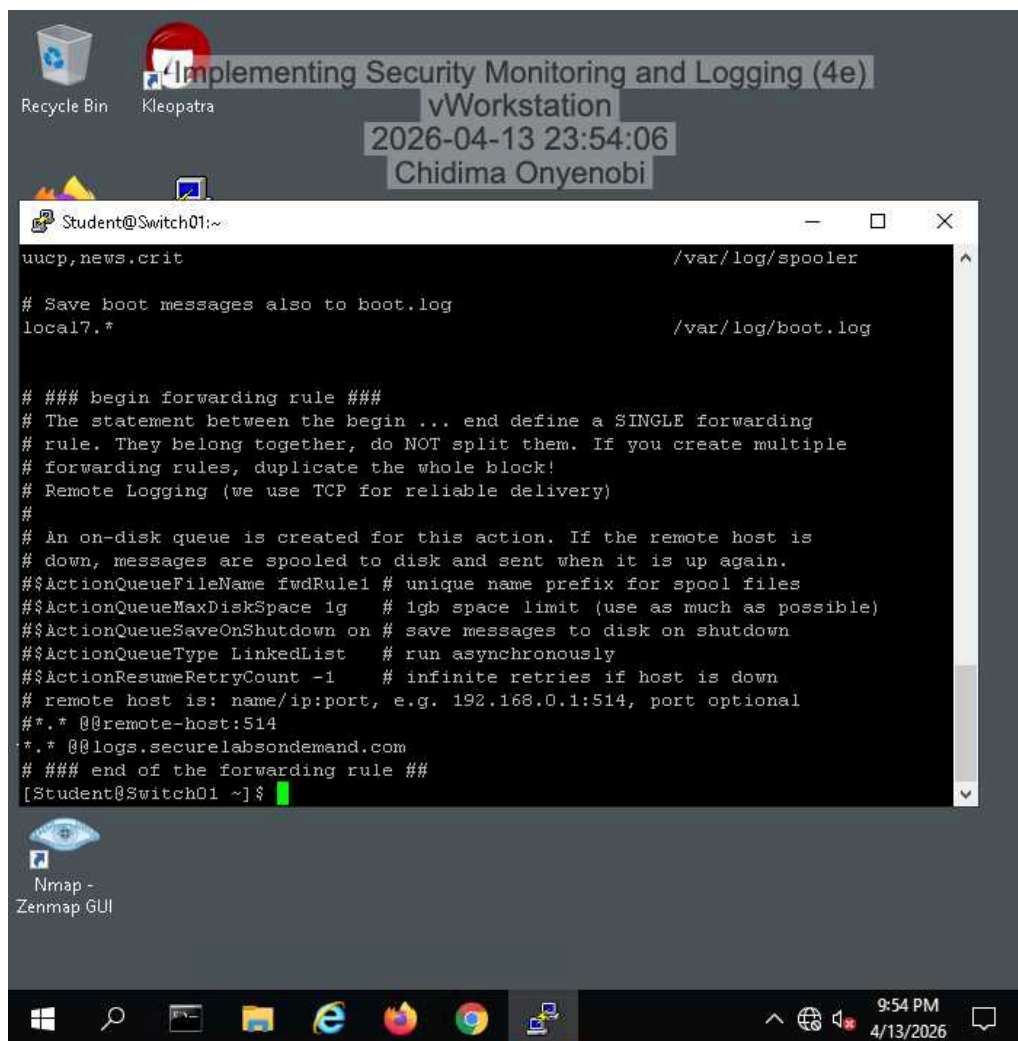
The screenshot shows a web browser window displaying the pfSense interface. The main content area shows the 'Active Log' for Snort, which contains 6 entries. Each entry is a row in a table with columns: Date, Action, Pri, Proto, Class, Source IP, SPort, Destination IP, and DPort. All entries are ICMP alerts with a priority of 3, classified as 'Misc activity'. The source IP is 172.31.0.1 and the destination IP is 172.30.0.2. The timestamps are 2026-04-14 04:16:05 and 04:16:06. A taskbar at the bottom shows the Windows logo, search icon, and several application icons. The system clock in the bottom right corner indicates 9:32 PM on 4/13/2026.

Date	Action	Pri	Proto	Class	Source IP	SPort	Destination IP	DPort
2026-04-14 04:16:07	⚠	3	ICMP	Misc activity	172.31.0.1		172.30.0.2	
2026-04-14 04:16:07	⚠	3	ICMP	Misc activity	172.31.0.1		172.30.0.2	
2026-04-14 04:16:06	⚠	3	ICMP	Misc activity	172.31.0.1		172.30.0.2	
2026-04-14 04:16:06	⚠	3	ICMP	Misc activity	172.31.0.1		172.30.0.2	
2026-04-14 04:16:05	⚠	3	ICMP	Misc activity	172.31.0.1		172.30.0.2	
2026-04-14 04:16:05	⚠	3	ICMP	Misc activity	172.31.0.1		172.30.0.2	

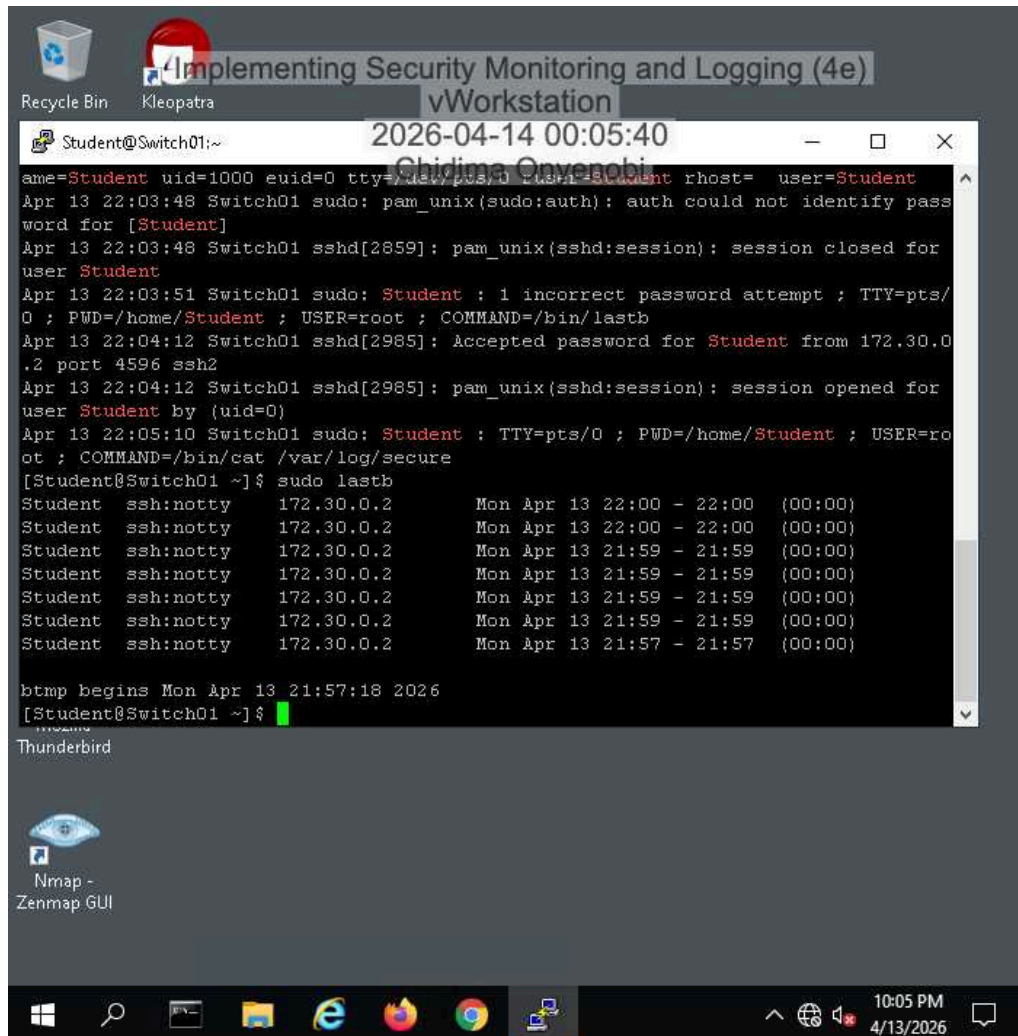
Applied Learning

Part 1: Identifying Failed Logon Attempts on Linux Systems

11. Make a screen capture showing the edited `rsyslog.conf` file.



22. Make a screen capture showing the failed login attempts.



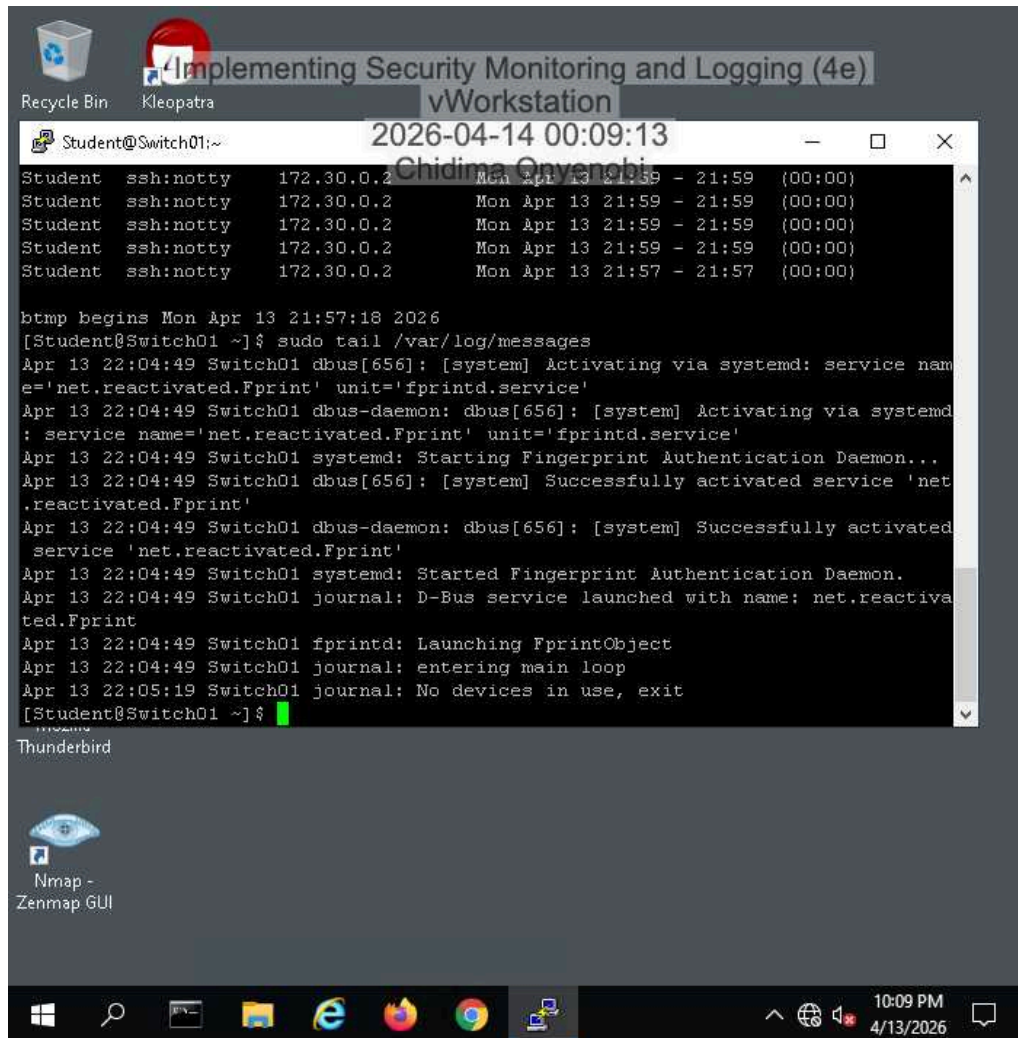
The screenshot shows a Windows desktop environment. At the top, there's a title bar for a window titled "Implementing Security Monitoring and Logging (4e)" with a timestamp of "2026-04-14 00:05:40". Below the title bar, a terminal window is open, displaying the following logs:

```
ame=Student uid=1000 euid=0 tty=/dev/pts/0 sshd=sshd[2985] rhost= user=Student
Apr 13 22:03:48 Switch01 sudo: pam_unix(sudo:auth): auth could not identify pass
word for [Student]
Apr 13 22:03:48 Switch01 sshd[2859]: pam_unix(sshd:session): session closed for
user Student
Apr 13 22:03:51 Switch01 sudo: Student : 1 incorrect password attempt ; TTY=pts/
0 ; PWD=/home/Student ; USER=root ; COMMAND=/bin/lastb
Apr 13 22:04:12 Switch01 sshd[2985]: Accepted password for Student from 172.30.0
.2 port 4596 ssh2
Apr 13 22:04:12 Switch01 sshd[2985]: pam_unix(sshd:session): session opened for
user Student by (uid=0)
Apr 13 22:05:10 Switch01 sudo: Student : TTY=pts/0 ; PWD=/home/Student ; USER=ro
ot ; COMMAND=/bin/cat /var/log/secure
[Student@Switch01 ~]$ sudo lastb
Student  ssh:notty      172.30.0.2      Mon Apr 13 22:00 - 22:00      (00:00)
Student  ssh:notty      172.30.0.2      Mon Apr 13 22:00 - 22:00      (00:00)
Student  ssh:notty      172.30.0.2      Mon Apr 13 21:59 - 21:59      (00:00)
Student  ssh:notty      172.30.0.2      Mon Apr 13 21:59 - 21:59      (00:00)
Student  ssh:notty      172.30.0.2      Mon Apr 13 21:59 - 21:59      (00:00)
Student  ssh:notty      172.30.0.2      Mon Apr 13 21:59 - 21:59      (00:00)
Student  ssh:notty      172.30.0.2      Mon Apr 13 21:57 - 21:57      (00:00)

btmp begins Mon Apr 13 21:57:18 2026
[Student@Switch01 ~]$
```

Below the terminal window, there's a taskbar with several icons: Recycle Bin, Kleopatra, Thunderbird, Nmap - Zenmap GUI, and a system tray showing the time as 10:05 PM on 4/13/2026.

24. Make a screen capture showing the last 10 log messages.



The screenshot shows a Windows desktop environment. At the top, there is a taskbar with icons for Recycle Bin, Kleopatra, and a window titled "Implementing Security Monitoring and Logging (4e) vWorkstation". The main window is a terminal window titled "Student@Switch01:~" with a timestamp of "2026-04-14 00:09:13". The terminal displays the following log messages:

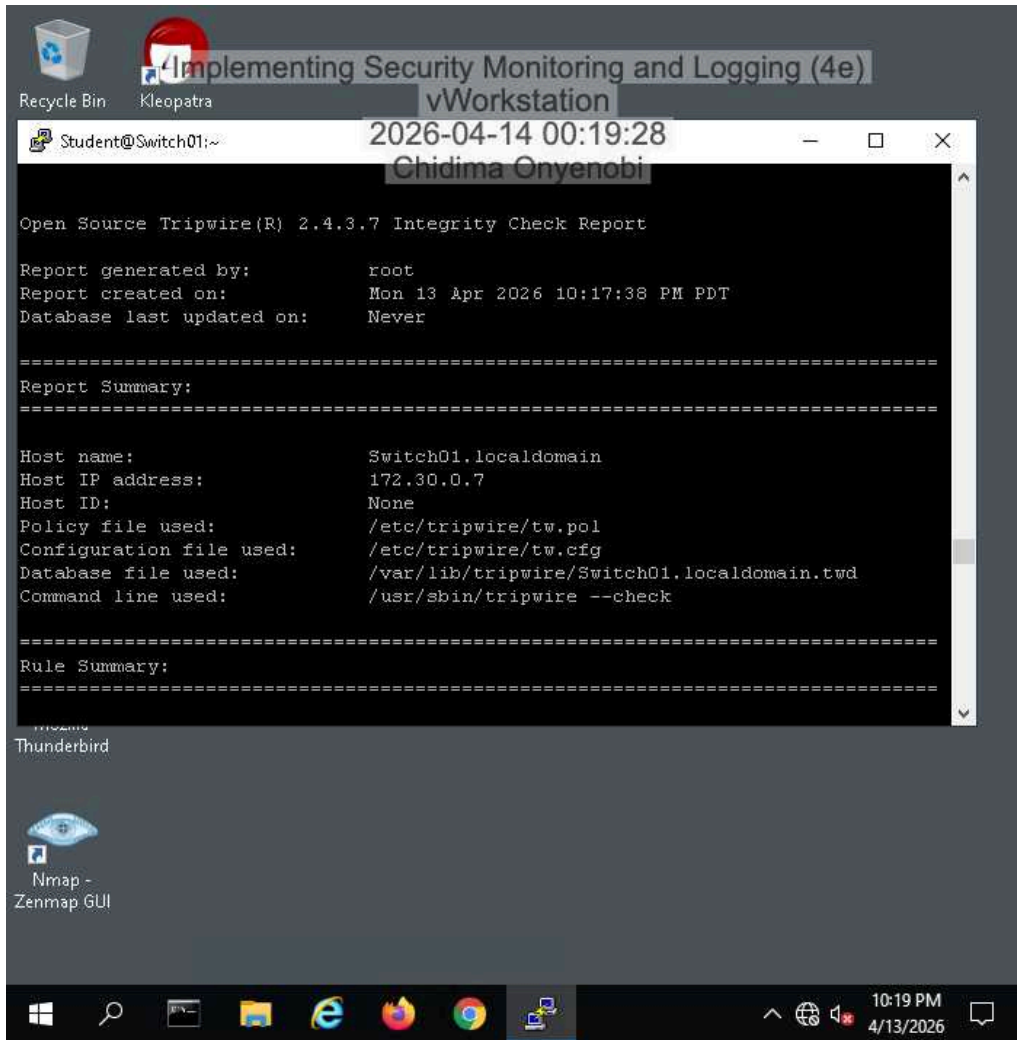
```
Student ssh:notty 172.30.0.2 Mon Apr 13 21:59 - 21:59 (00:00)
Student ssh:notty 172.30.0.2 Mon Apr 13 21:59 - 21:59 (00:00)
Student ssh:notty 172.30.0.2 Mon Apr 13 21:59 - 21:59 (00:00)
Student ssh:notty 172.30.0.2 Mon Apr 13 21:59 - 21:59 (00:00)
Student ssh:notty 172.30.0.2 Mon Apr 13 21:57 - 21:57 (00:00)

btmtp begins Mon Apr 13 21:57:18 2026
[Student@Switch01 ~]$ sudo tail /var/log/messages
Apr 13 22:04:49 Switch01 dbus[656]: [system] Activating via systemd: service nam
e='net.reactivated.Fprint' unit='fprintd.service'
Apr 13 22:04:49 Switch01 dbus-daemon: dbus[656]: [system] Activating via systemd
: service name='net.reactivated.Fprint' unit='fprintd.service'
Apr 13 22:04:49 Switch01 systemd: Starting Fingerprint Authentication Daemon...
Apr 13 22:04:49 Switch01 dbus[656]: [system] Successfully activated service 'net
.reactivated.Fprint'
Apr 13 22:04:49 Switch01 dbus-daemon: dbus[656]: [system] Successfully activated
service 'net.reactivated.Fprint'
Apr 13 22:04:49 Switch01 systemd: Started Fingerprint Authentication Daemon.
Apr 13 22:04:49 Switch01 journal: D-Bus service launched with name: net.reactiva
ted.Fprint
Apr 13 22:04:49 Switch01 fprintd: Launching FprintObject
Apr 13 22:04:49 Switch01 journal: entering main loop
Apr 13 22:05:19 Switch01 journal: No devices in use, exit
[Student@Switch01 ~]$
```

Below the terminal window, there are icons for Thunderbird, Nmap - Zenmap GUI, and a taskbar at the bottom showing the Windows logo, search icon, and system clock (10:09 PM 4/13/2026).

Part 2: Monitoring File Integrity with Tripwire

12. Make a screen capture showing the **Object Summary** section for the Tripwire report.



The screenshot shows a Windows desktop environment. In the background, there is a window titled "Implementing Security Monitoring and Logging (4e) vWorkstation" with a timestamp of "2026-04-14 00:19:28" and the name "Chidima Onyenobi". The desktop has icons for "Recycle Bin", "Kleopatra", "Thunderbird", and "Nmap - Zenmap GUI". The taskbar at the bottom shows the Start button, search icon, and several application icons. A terminal window is open, displaying the output of a Tripwire integrity check. The terminal text is as follows:

```
Open Source Tripwire(R) 2.4.3.7 Integrity Check Report

Report generated by:      root
Report created on:       Mon 13 Apr 2026 10:17:38 PM PDT
Database last updated on: Never

=====
Report Summary:
=====

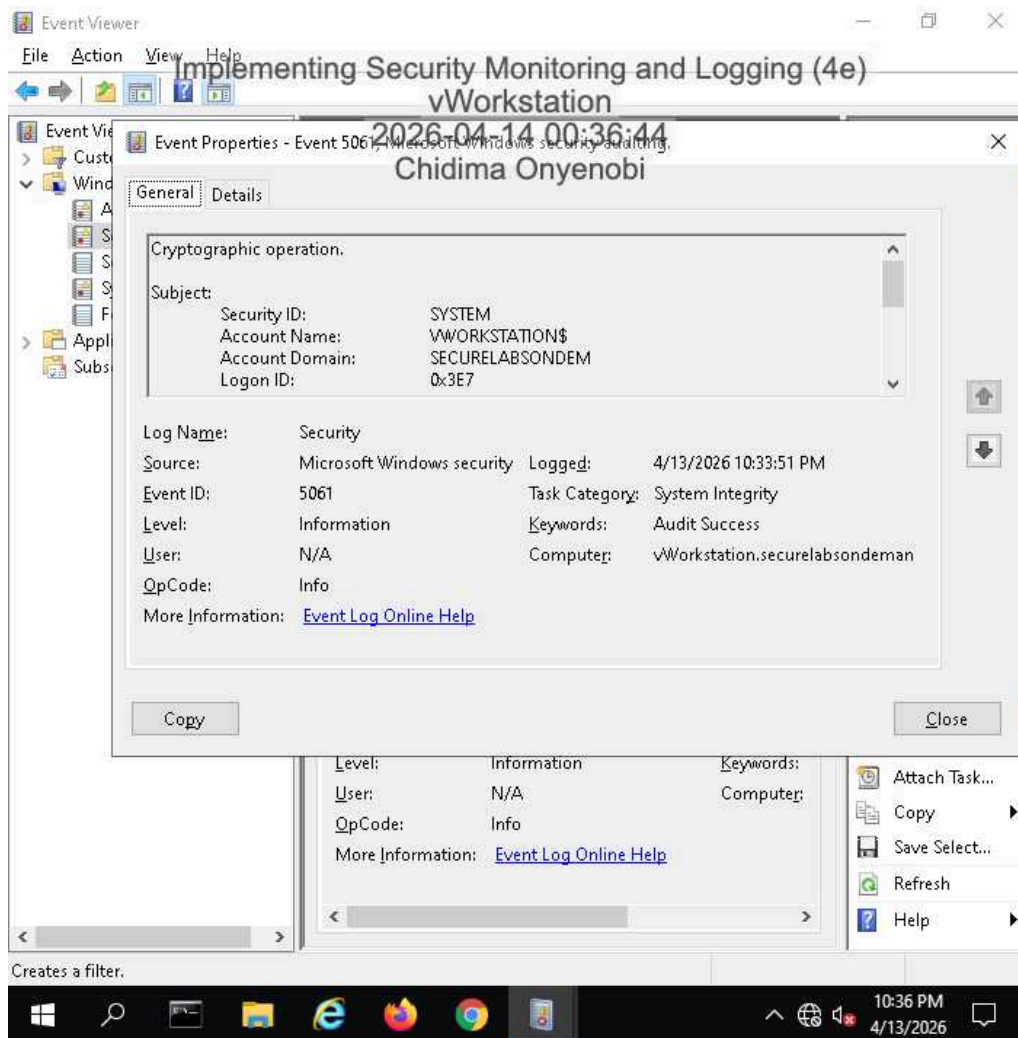
Host name:                Switch01.localdomain
Host IP address:          172.30.0.7
Host ID:                  None
Policy file used:         /etc/tripwire/tw.pol
Configuration file used:  /etc/tripwire/tw.cfg
Database file used:       /var/lib/tripwire/Switch01.localdomain.twd
Command line used:        /usr/sbin/tripwire --check

=====
Rule Summary:
=====
```

Challenge and Analysis

Part 1: Identifying Additional Event Types in the Event Viewer

Make a screen capture showing the **Security Event Properties** dialog box for an **Audit Failure** associated with **Event ID 5061**.



Provide a brief explanation of the operation that would generate a security event with Event ID 5061.

The operation that would generate a security event with Event ID 5061 is a cryptographic action occurring within the Windows security system.

Part 2: Configuring Snort as an Intrusion Prevention System

Implementing Security Monitoring and Logging (4e)

Fundamentals of Information Systems Security, Fourth Edition - Lab 08

Make a screen capture showing the **Legacy Blocking Mode** enabled on the LAN interface.

